

Text Document

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```
from flask import Flask, render_template, request
import pandas as pd
import joblib
```

```
app = Flask(__name__)
```

```
# Load Model
```

```
model = joblib.load("flood_model.pkl")
```

```
@app.route("/")
```

```
def home():
```

```
    return render_template(
```

```
        "index.html",
```

```
        prediction="",
```

```
        rainfall=0,
```

```
        deaths=0,
```

```
        affected=0,
```

```
        risk=0
```

```
    )
```

```
@app.route("/predict", methods=["POST"])
```

```
def predict():
```

```
    try:
```

```
        disaster_type = int(request.form.get("Disaster_Type", 0))
```

```
        latitude = float(request.form.get("Latitude", 0))
```

```
        longitude = float(request.form.get("Longitude", 0))
```

```
        total_deaths = float(request.form.get("Total_Deaths", 0))
```

Text Document

```
total_affected = float(request.form.get("Total_Affected", 0))
duration = float(request.form.get("duration", 0))
time = float(request.form.get("time", 0))
rainfall = float(request.form.get("Rainfall", 0))
elevation = float(request.form.get("Elevation", 0))
slope = float(request.form.get("Slope", 0))
distance = float(request.form.get("distance", 0))
```

```
df = pd.DataFrame([{
    "Disaster Type": disaster_type,
    "Latitude": latitude,
    "Longitude": longitude,
    "Total Deaths": total_deaths,
    "Total Affected": total_affected,
    "duration": duration,
    "time": time,
    "Rainfall": rainfall,
    "Elevation": elevation,
    "Slope": slope,
    "distance": distance
}])
```

```
prediction = model.predict(df)[0]
```

```
if hasattr(model, "predict_proba"):
    risk = int(model.predict_proba(df)[0][1] * 100)
else:
    risk = 85 if prediction == 1 else 15
```

```
if prediction == 1:
    result = "Ø<ß Flood Likely"
else:
    result = " No Flood "
```

```
return render_template(
    "index.html",
    prediction=result,
    rainfall=rainfall,
    deaths=total_deaths,
    affected=total_affected,
    risk=risk
)
```

```
except Exception as e:
    return render_template(
        "index.html",
        prediction="Error: " + str(e),
        rainfall=0,
        deaths=0,
        affected=0,
        risk=0
    )
```

```
if __name__ == "__main__":
    app.run(debug=True)
```